# Did Streaming Break How Songs Are Built?

A Multi-Method Analysis of Structural Changes in Music Composition Across the Streaming Era

DS108: Introduction to Data Analysis with Python

# 01 Introduction

Research Questions, Hypotheses, Prior Work & Techniques

# Research Questions

**Main: Did the shift to algorithmic streaming (post-2018) structurally change song composition compared to the pre-streaming era (2000-2009)?**

### Branch 1 — Loudness-Valence Trade-off

Has the systemic engineering of track loudness degraded the emotional positivity (valence) of music?
Method: Pearson's Correlation

### Branch 2 — Tempo Convergence

Has popular music converged toward a narrower tempo range over time?
Method: Variance Comparison + Levene's Test

# Hypotheses

### H1 Duration

Tracks post-2018 have significantly lower mean duration than tracks from 2000-2009.

Welch's t-test

### H2 Tempo

Tracks post-2018 have significantly higher mean tempo than tracks from 2000-2009.

Welch's t-test

### H3 Loudness-Valence

Significant negative correlation between loudness and valence — louder tracks are less happy.

Pearson's r

### H4 Tempo Convergence

Post-2015 tracks show lower tempo variance than earlier decades, indicating BPM standardization.

Levene's Test

# Previous Research

- The "Loudness War" — mastering levels have increased steadily since the 1990s as producers compress dynamic range to make tracks louder on radio and streaming (Deruty & Tardieu, 2014).
- Studies using the Million Song Dataset showed trends toward decreasing timbral diversity and increasing loudness over decades (Serra et al., 2012).
- Spotify's recommendation algorithm favors features like danceability, energy, and track duration for playlist placement (Anderson et al., 2020).
- Observational analyses show average track length declining from ~4:00 to ~3:30 since 2017, coinciding with the streaming revenue model shift (Spotify Insights).

# Statistical Techniques Utilized

| Technique | Purpose | Application |
|---|---|---|
| Descriptive Statistics | Centrality & dispersion | Mean, median, SD, IQR |
| IQR Outlier Detection | Identify anomalies | Cap extreme values |
| Z-score Standardization | Normalize variables | Correlation analysis prep |
| Shapiro-Wilk Test | Check normality | Subsamples (n=5,000) |
| Kolmogorov-Smirnov | Supplementary normality | Alongside Shapiro-Wilk |
| Levene's Test | Homogeneity of variances | Before t-tests & variance comp. |
| Welch's t-test | Compare two group means | Duration & tempo across eras |
| Cohen's d | Practical effect size | Supplement p-values |
| Pearson's r | Linear relationship | Loudness vs. valence |
| Fisher Z-transform | Compare two correlations | Era correlation comparison |
| Spearman's rho | Monotonic trend | Year vs. tempo variance |

# 02 Method

Dataset, Descriptive Statistics & Preprocessing Pipeline

# Dataset Description

| 1,048,575 | 82 | 22 | 19 |
|---|---|---|---|
| Total Tracks | Genres | Years (2000-2023) | Variables |

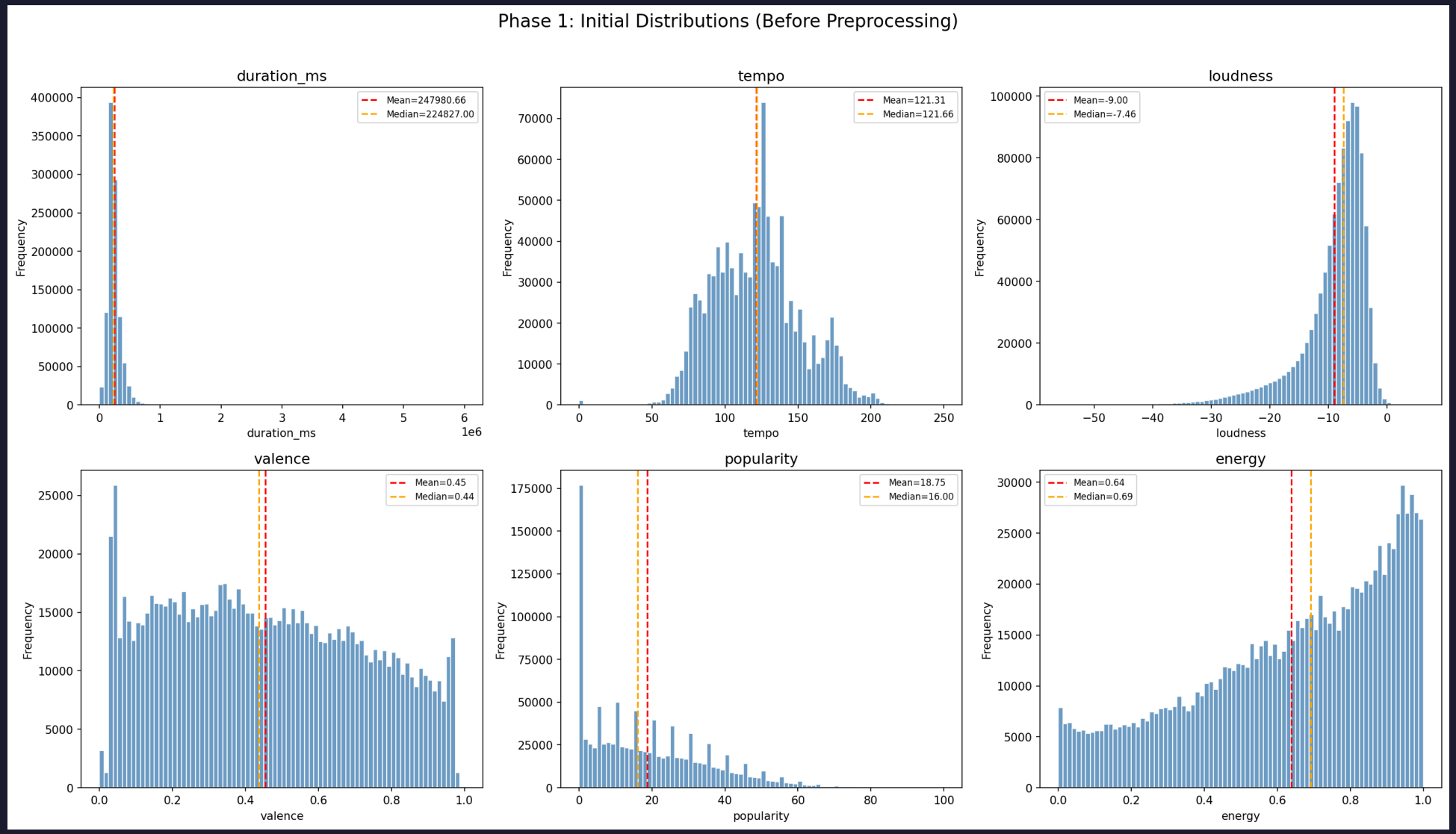
| Group | Years | Tracks |
|---|---|---|
| Pre-streaming | 2000-2009 | 425,924 |
| Transition | 2012-2017 | 309,313 |
| Streaming era | 2018-2023 | 313,338 |

ℹ Source: Spotify audio features dataset | Missing values: 0 | Duplicates: 0

# Descriptive Statistics — Before Preprocessing

| Variable | Mean | Median | SD | Min | Q1 | Q3 | IQR |
|---|---|---|---|---|---|---|---|
| popularity | 18.75 | 16.00 | 16.07 | 0 | 5.0 | 29.0 | 24.0 |
| danceability | 0.538 | 0.551 | 0.185 | 0.0 | 0.413 | 0.678 | 0.265 |
| energy | 0.639 | 0.692 | 0.271 | 0.0 | 0.452 | 0.872 | 0.420 |
| loudness | -9.00 | -7.46 | 5.71 | -56.1 | -10.84 | -5.29 | 5.56 |
| valence | 0.454 | 0.436 | 0.268 | 0.0 | 0.225 | 0.672 | 0.447 |
| tempo | 121.31 | 121.66 | 29.79 | 0.0 | 98.64 | 139.89 | 41.25 |
| duration_ms | 247,981 | 224,827 | 144,641 | 2,073 | 180,392 | 285,173 | 104,782 |

# Descriptive Statistics — Before Preprocessing



# Preprocessing Pipeline

Step 1

## Missing Values

No missing values found across all 19 columns. No imputation was required.

Step 2

## Outlier Detection (IQR Method)

Winsorization: cap extreme values at Q1-1.5xIQR and Q3+1.5xIQR boundaries.

Step 3

## Z-score Standardization

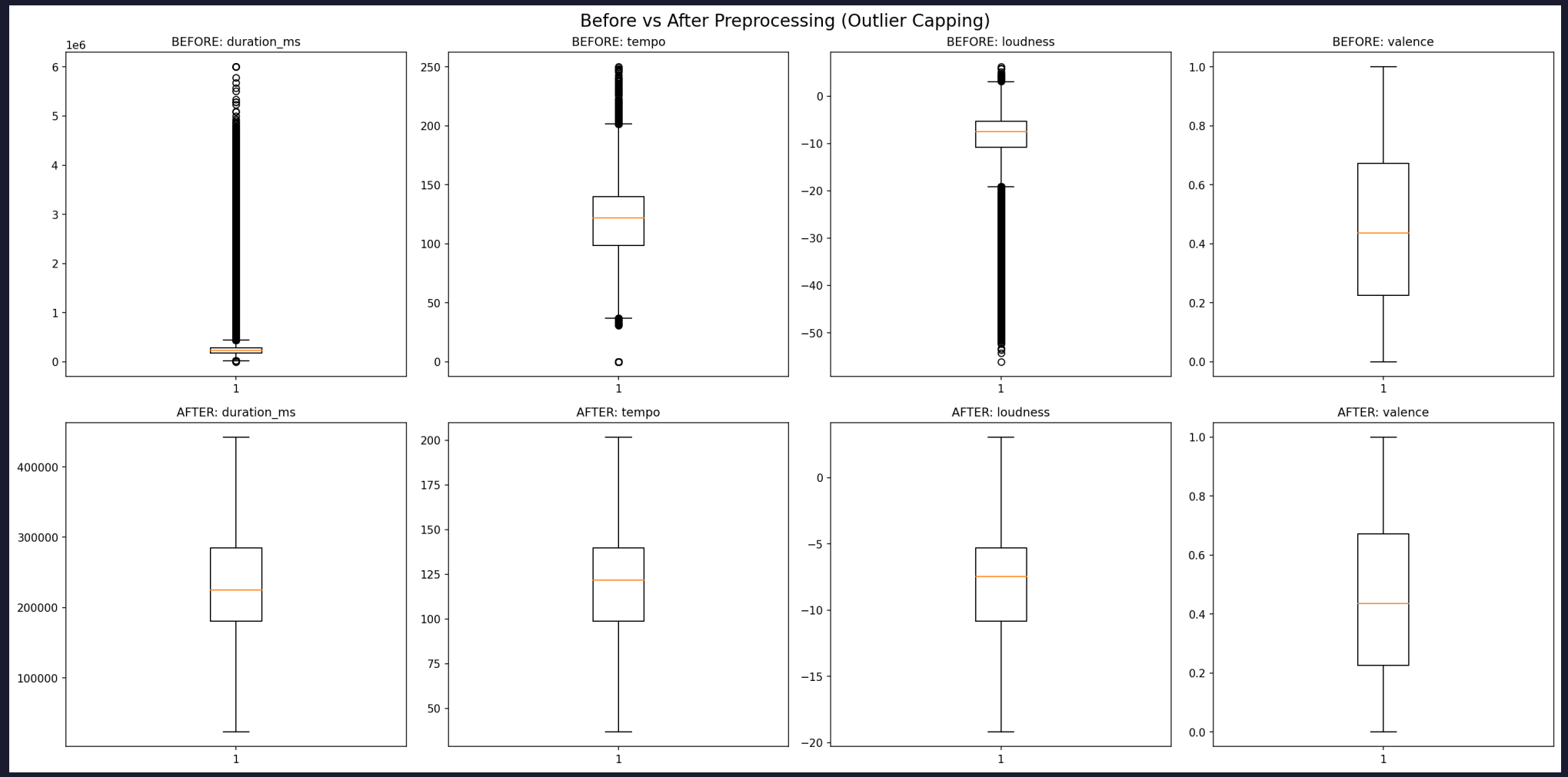
Standardize loudness, valence, tempo, duration_ms to mean=0, SD=1.

| Variable | Lower Bound | Upper Bound | Outliers | % of Data |
|---|---|---|---|---|
| duration_ms | 23,219 ms | 442,345 ms | 54,834 | 5.23% |
| tempo | 36.76 BPM | 201.77 BPM | 4,951 | 0.47% |
| loudness | -19.18 dB | 3.05 dB | 68,835 | 6.56% |
| valence | -0.45 | 1.34 | 0 | 0.00% |

# After Preprocessing — Comparison

| Variable | Before Mean | After Mean | Before SD | After SD |
|---|---|---|---|---|
| loudness | -9.00 dB | -8.62 dB | 5.71 | 4.55 |
| tempo | 121.31 BPM | 121.33 BPM | 29.79 | 29.61 |
| duration_ms | 247,981 ms | 238,864 ms | 144,641 | 89,518 |
| valence | 0.454 | 0.454 | 0.268 | 0.268 |

# After Preprocessing — Comparison



# 03 Results

T-tests, Correlation Analysis & Variance Comparison

# Assumption Checks

## Normality — Shapiro-Wilk (n=5,000 subsamples)

| Variable | Group | W | p-value |
|---|---|---|---|
| duration_ms | Pre-streaming | 0.9652 | 4.31e-33 |
| duration_ms | Streaming era | 0.9469 | 3.66e-39 |
| tempo | Pre-streaming | 0.9852 | 1.76e-22 |
| tempo | Streaming era | 0.9900 | 2.24e-18 |

All tests reject normality ($p < 0.001$). Expected with $n > 400{,}000$ per group.
By the Central Limit Theorem, the t-test remains robust at these sample sizes.

## Homoscedasticity — Levene's Test

| Variable | F | p-value | Result |
|---|---|---|---|
| duration_ms | 8,271.33 | < 2.2e-308 | Unequal variances |
| tempo | 4.09 | 4.30e-02 | Unequal variances |

→ **Using Welch's t-test (does not assume equal variances)**

# H1 Result: Song Duration — SUPPORTED ✓

### Pre-Streaming (2000-2009)

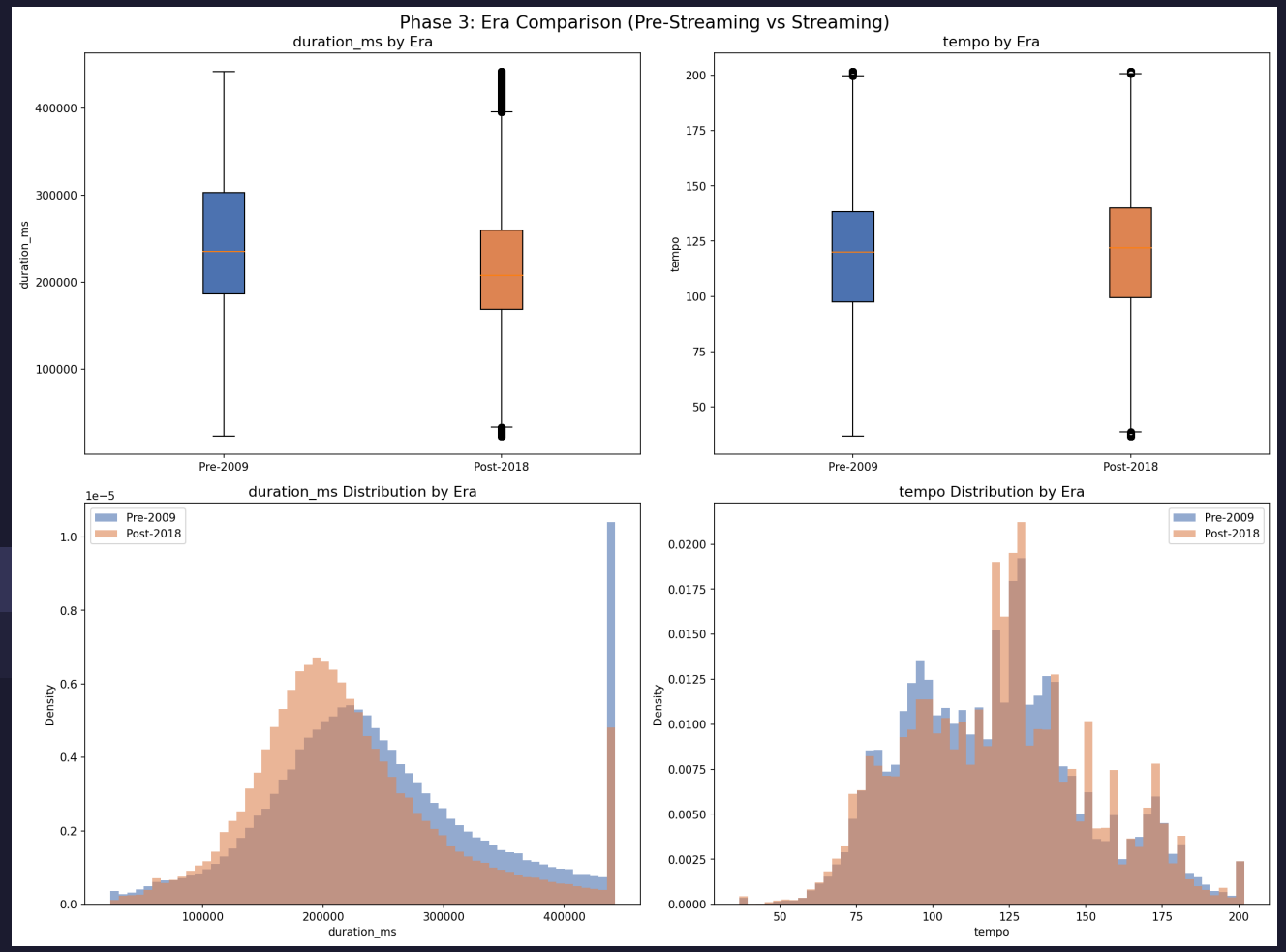
Mean: 249,304 ms (4:09) | Median: 3:55
SD: 94,358 ms | n = 425,924

### Streaming Era (2018-2023)

Mean: 221,489 ms (3:41) | Median: 3:28
SD: 81,964 ms | n = 313,338

| Welch's t | p-value | Cohen's d | Mean diff |
|---|---|---|---|
| 135.17 | < 2.2e-308 | 0.315 (small) | -28 sec (-11.16%) |

Songs lost ~28 seconds on average — nearly half a verse. The median dropped from 3:55 to 3:28, confirming this is a population-level shift, not outlier-driven.



# H2 Result: Tempo — NEGLIGIBLE EFFECT

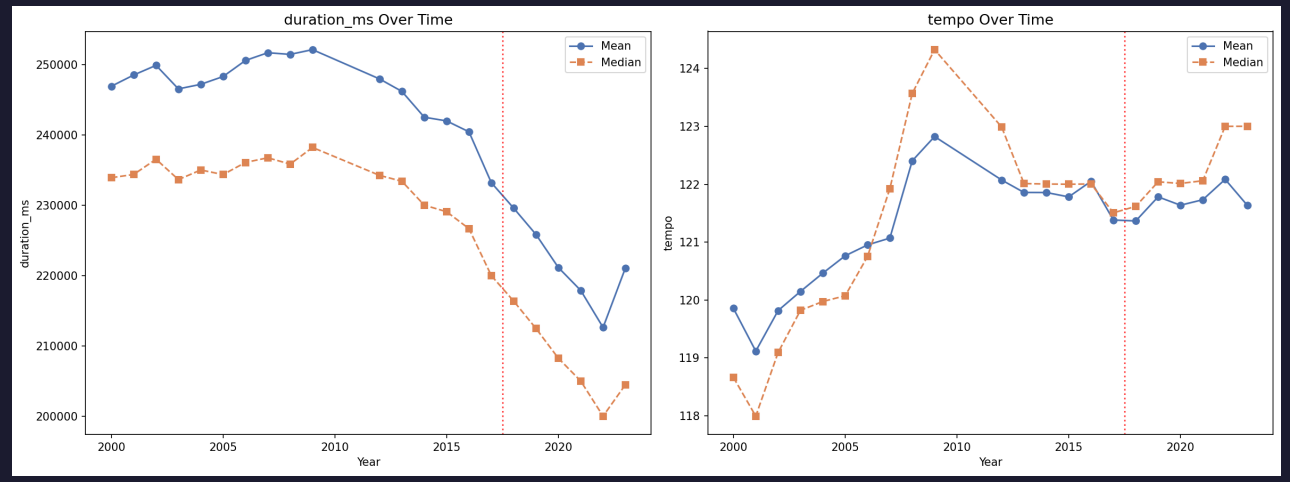
### Pre-Streaming

Mean: 120.70 BPM | SD: 29.59 | n = 425,924

### Streaming Era

Mean: 121.71 BPM | SD: 29.78 | n = 313,338

| Welch's t | p-value | Cohen's d | Mean diff |
|---|---|---|---|
| -14.45 | 1.28e-47 | 0.034 (negligible) | +1.01 BPM (+0.84%) |




Statistically significant ($p < 0.001$) but practically meaningless — only +1 BPM.

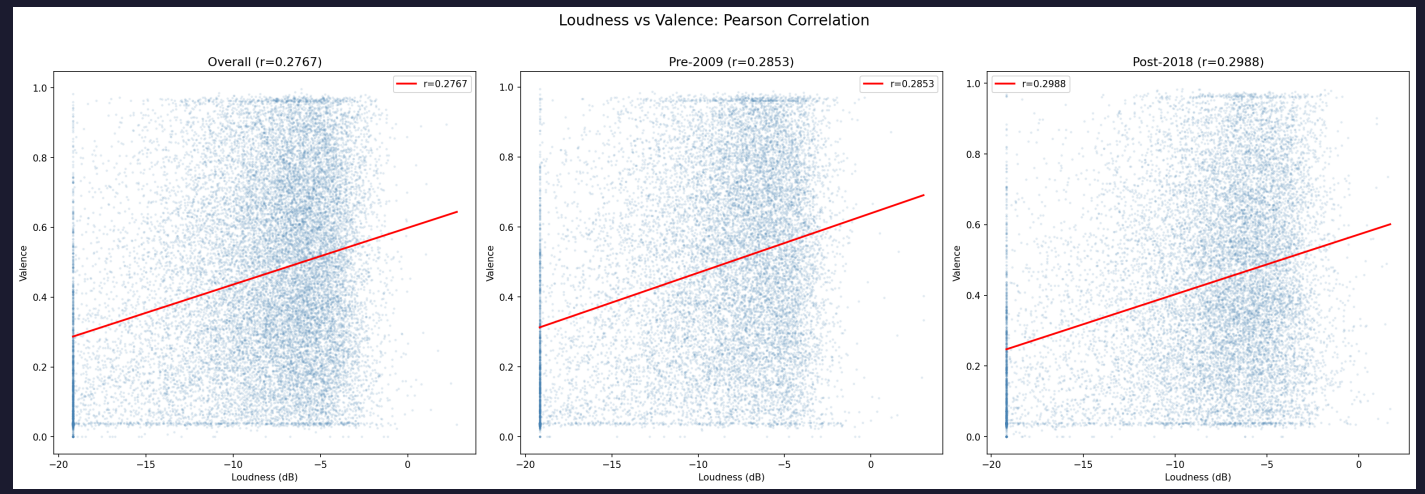
A textbook example of how massive sample sizes make trivial differences "significant." Streaming has NOT made music faster.

# H3 Result: Loudness-Valence — REJECTED (Reversed!)

## Overall Correlation

Pearson's r = +0.277
p-value < 2.2e-308
$r^2$ = 0.077

| Era | Pearson's r | $r^2$ | n |
|---|---|---|---|
| Pre-streaming | +0.285 | 0.081 | 425,924 |
| Streaming era | +0.299 | 0.089 | 313,338 |




Hypothesis predicted a negative correlation. Data shows the opposite: louder = happier.

Explanation: high-energy genres (pop, EDM) are both loud and positive. Quiet genres (ambient, classical) are softer and sadder.

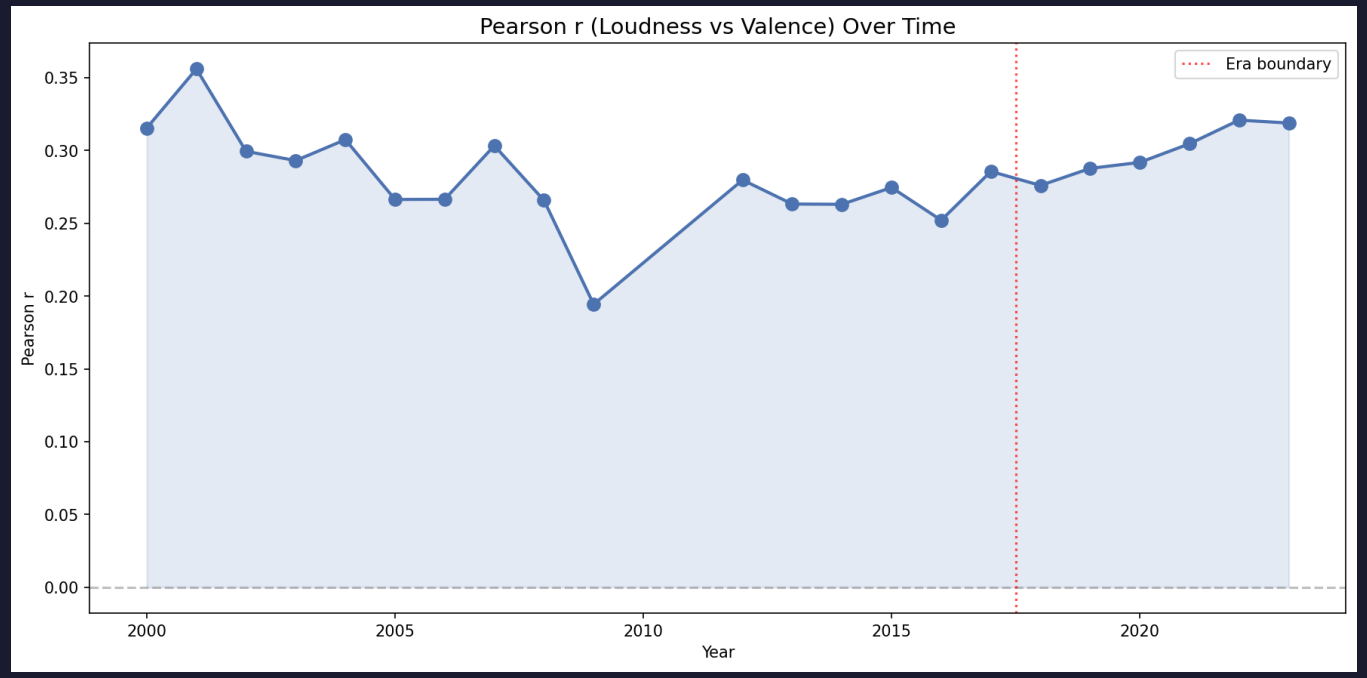
Fisher Z-test: correlation has significantly strengthened in the streaming era ($p < 0.001$).

# Loudness-Valence Coupling Over Time

The correlation dipped to r = 0.194 in 2009 and has climbed steadily since 2016.

Recent highs of r = 0.321 (2022) represent the strongest coupling in over a decade.

This may reflect increasing genre polarization in streaming playlists — sorting music into neat emotional boxes.

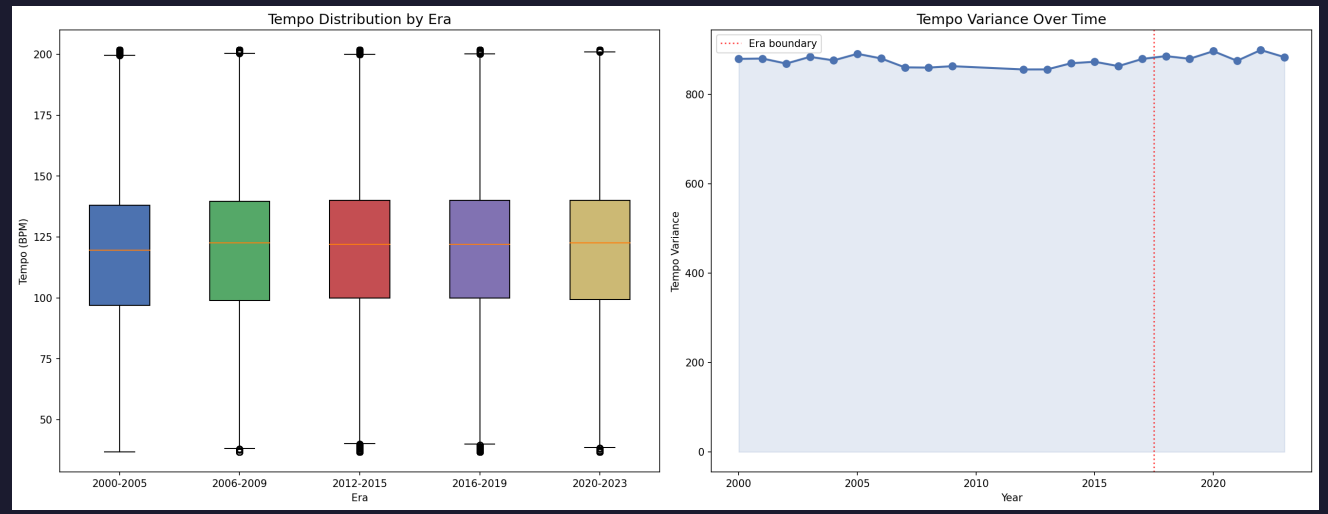



| Metric | Value |
|---|---|
| Fisher Z-test | -6.25 |
| p-value | 4.16e-10 |
| Result | Significant difference |

# H4 Result: Tempo Convergence — REJECTED

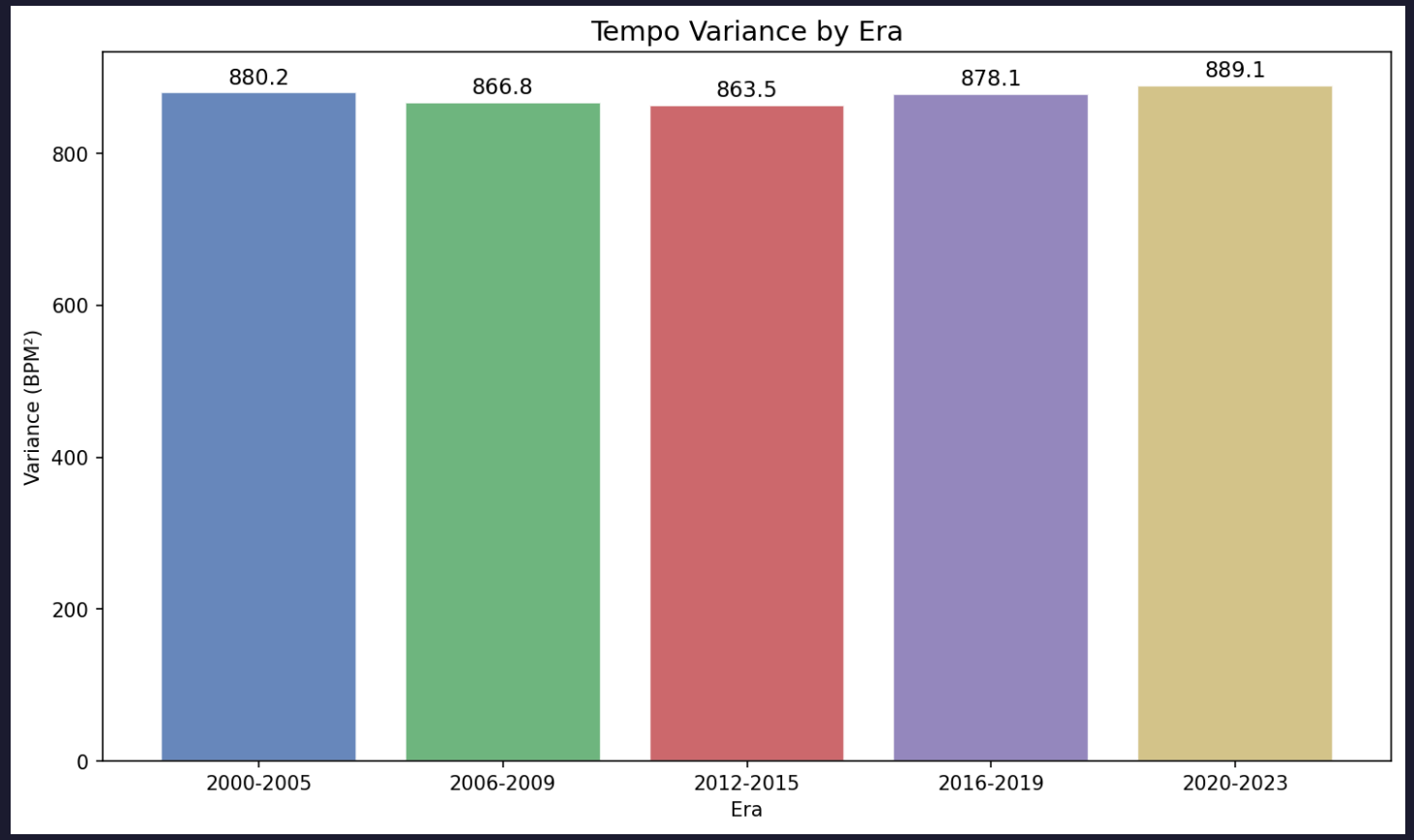
| Era | Variance | SD | IQR |
|---|---|---|---|
| 2000-2005 | 880.23 | 29.67 | 41.08 |
| 2006-2009 | 866.79 | 29.44 | 40.59 |
| 2012-2015 | 863.46 | 29.38 | 39.99 |
| 2016-2019 | 878.09 | 29.63 | 40.10 |
| 2020-2023 | 889.11 | 29.82 | 40.66 |

| Test | Statistic | p-value |
|---|---|---|
| Levene's (omnibus) | F = 53.33 | 5.15e-45 |
| Spearman trend | ρ = 0.203 | 0.366 (n.s.) |




Variance follows a U-shape, not a decline.
Lowest point: 2012-2015 (peak EDM era).
Highest point: 2020-2023 — streaming may actually be diversifying tempo via long-tail.

# Tempo Variance by Era




Key Pairwise Comparisons

2000-05 vs 2012-15: F=189, p<0.001 ✓

2006-09 vs 2016-19: F=1.58, p=0.208 ✗
(only non-significant pair)

2012-15 vs 2020-23: F=76.4, p<0.001 ✓

Variance rebounded after the
EDM-era minimum.

# 04 Discussion

Interpretation, Validity, Limitations & Future Directions

# Summary of Findings

| Hypothesis | Supported? | Key Finding |
|---|---|---|
| H1: Songs shorter post-2018 | YES | -28 sec (-11.16%), d = 0.315 |
| H2: Songs faster post-2018 | NO (negligible) | +1 BPM (+0.84%), d = 0.034 |
| H3: Louder = less happy | NO (reversed) | r = +0.277, louder = happier |
| H4: Tempo converging | NO (U-shaped) | Variance highest in 2020-2023 |

Streaming has measurably shortened songs — the single strongest finding. But tempo, emotional tone, and rhythmic variety have remained resilient. The industry adapted to streaming economics by trimming length, while the deeper musical DNA of songs has been remarkably unchanged.

# Validity & Reliability

## Internal Validity

Massive dataset (>1M tracks) reduces sampling error.

Welch's t-test accounts for unequal variances. Effect sizes supplement p-values.

Cannot establish causation (observational design).

## External Validity

82 genres, 22 years — broad cross-section of music.

Data is Spotify-only, may over-represent Western pop and under-represent local or underground scenes.

## Construct Validity

"Valence" is algorithmic, not a direct measure of listener emotional experience.

"Loudness" is objective (LUFS).

"Tempo" can be confused by half-time / double-time.

## Reliability

Internal consistency: Spotify features are algorithmically computed — single measurement system.
Test-retest: Audio features are computed once at upload — inherently stable over time.
Cross-validation: Not applicable to this descriptive/correlational design.

# Limitations & Flaws

- Correlation, not causation — cannot prove streaming caused shorter songs. Other factors (TikTok, attention spans, smartphones) may be responsible.
- Survivorship bias — only tracks available on Spotify; delisted or region-locked tracks are absent.
- Genre imbalance — some genres have more tracks than others, which may skew aggregate statistics.
- 2010-2011 gap — missing years create a discontinuity in the timeline.
- Popularity score dynamics — Spotify's metric is time-decayed and recency-biased; older hits appear less popular.
- Ecological fallacy — aggregate trends across all genres may not hold within specific genres.

# Future Directions

### Within-genre analysis

Test whether the duration decline is universal or specific to genres like pop and hip-hop.

### Causal inference

Use interrupted time-series or regression discontinuity around streaming milestones.

### Intro length analysis

Test if the first 30 seconds have become more hook-dense as a response to skip algorithms.

### Cross-platform comparison

Compare Spotify features with Apple Music and YouTube data for external validation.

### Listener behavior data

Pair audio features with skip rates and completion rates for behavioral evidence.

# Conclusion

Streaming has made songs shorter — by an average of 28 seconds (11%) — but it hasn't made them faster, sadder, or more uniform in tempo.

The music industry adapted to streaming economics in the most direct way possible: by trimming track length. Revenue-per-stream (not per-minute) incentivizes more songs and shorter plays.

But the deeper musical DNA — emotional tone, rhythmic variety, loudness-mood coupling — has remained remarkably resilient.

Where we predicted homogenization, we found either no effect (tempo) or the opposite of our hypothesis (loudness-valence). The streaming era has shortened the container, but not changed what's inside it.

# Thank You

Questions?

Data source: Spotify Audio Features Dataset | 1,048,575 tracks | 2000-2023
Analysis: Python (pandas, scipy, matplotlib, seaborn)